# CS 401: HOMEWORK 3 GRADING RUBRIC & BENCHMARK SOLUTIONS

Lead TA: Maya Lin | Course Instructor: Dr. Arthur Chen | Total Points: 100

## Problem Breakdown & Point Allocation

Question 1: Recurrence Relation & Dynamic Programming Formulation (25 pts)
- Correct state definition and base cases: 10 pts
- Optimal substructure proof & recurrence: 10 pts
- Space-optimized implementation O(N): 5 pts

Question 2: Graph Shortest Path & Bounded Constraint Pruning (35 pts)
- Dijkstra state augmentation: 15 pts
- Handling negative edge boundary conditions: 10 pts
- Complexity analysis: 10 pts

Question 3: Multimodal Feature Projection & Contrastive Loss (40 pts)
- Mathematical derivation of InfoNCE denominator: 15 pts
- Numerical stability checks (log-sum-exp trick): 15 pts
- Experimental training curve evaluation: 10 pts

## Late Penalty & Regrade Policy

Students have 7 calendar days after grade release to submit regrade requests via Gradescope.